RNA-Seq Analysis – GSE288148

This report is for the analysis of the RNA-seq dataset GSE288148, which explores tumour-immune interactions in Mus musculus using CD8+ T cell populations isolated from tumour-neighbouring and distal regions. The data was processed using the RNA-seq analysis pipeline, including quality control, alignment, normalisation, differential expression, and functional enrichment analysis.

Quality Control

General stats from the Multiqc report for GSE288148 suggest that the dataset consists of paired-end reads without low-quality read files provided. Alignment to the genome was consistently high across all six samples (>90%), indicating very good data quality. The sequence duplication rate ranges between 60% and 75%, which is relatively high, but this is typical for RNA-seq data and may also reflect how Fastqc handles counting for paired-end libraries. Given that the other metrics were all acceptable, no trimming or elimination steps were performed, and the data were considered suitable to continue with alignment and RNA-seq analysis.

Quality control results and their interpretation were based on outputs from Multiqc and guidance adapted from the ICA practical document "Assessing Sequence and Alignment Quality with Multiqc", as discussed during class.

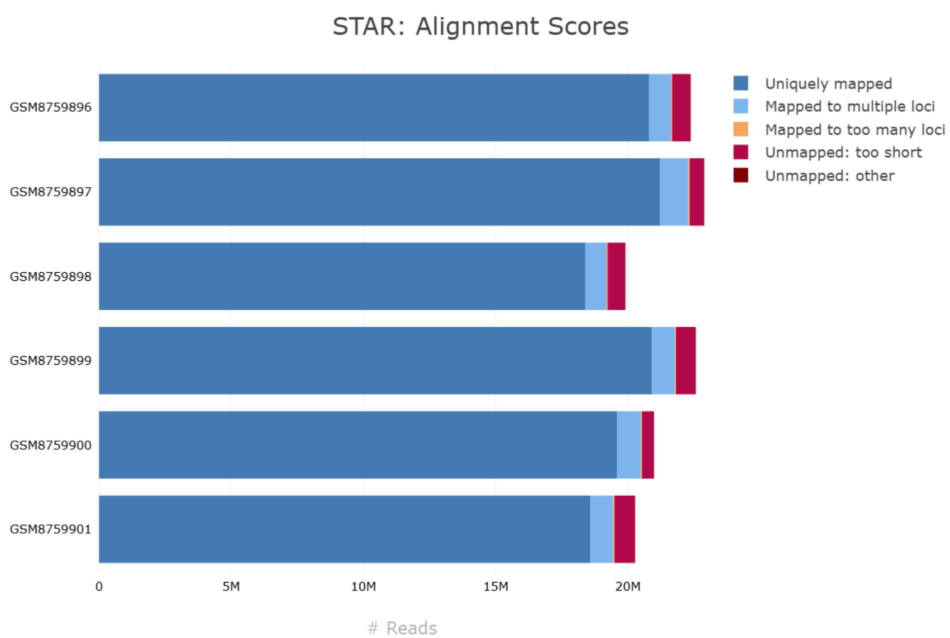

Figure 1: STAR alignment results for all six samples. The majority of reads in each library map uniquely to the genome. Rare occurrences of multimapping and unmapped reads indicate high-quality libraries suitable for downstream analysis, such as differential expression.

For these libraries, most sequences were uniquely aligned to the reference genome. Very few reads were mapped to multiple loci or failed to map entirely, further indicating that the samples were of sufficient quality for further analysis.

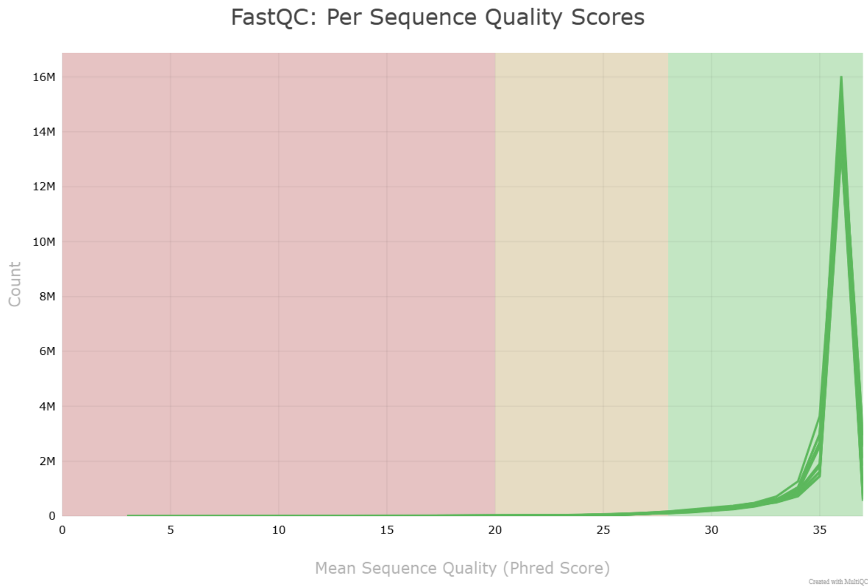

Figure 2: Per sequence quality scores across samples. The majority of reads fall within high Phred quality ranges, with small variability between samples. This indicates a strong, consistent base-level sequencing quality across all libraries.

Average Phred scores for reads are all in the high Phred score region, and there are no major differences between the sequences, again indicating high-quality reads for this analysis.

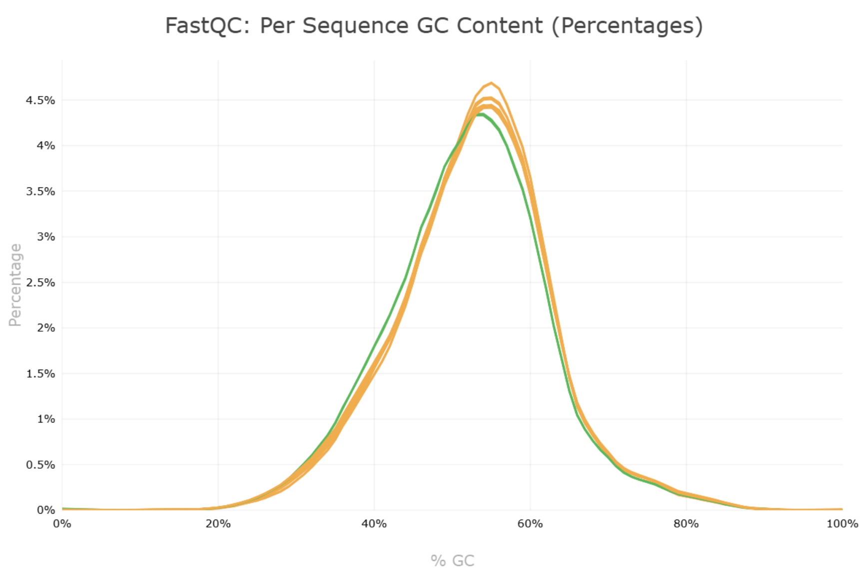

Figure 3: GC content distribution plot. The GC content is centred around ~50% for all samples. Some minor deviations (highlighted in yellow) were observed, but do not indicate any major bias in the dataset.

GC content seems mostly normally distributed with some sequences showing deviations (yellow colored distributions), however, they are distributed around ~50%, which is consistent with expectations for Mus musculus. Hence, no major bias was detected for the datasets.

The sequence length distribution section stated that the length was uniform across samples, with all reads being 150 bp in length, as expected from Illumina sequencing (confirmed from GEO record GSE288148, https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE288148).

Adapter content showed a very slight increase toward the end of read positions in the samples, but this was minimal and not considered significant. Therefore, no adapter trimming was required.

There were no overrepresented sequences in any of the libraries. This suggests that the reads are not affected by contamination, PCR bias, or leftover adapter content in a way that would interfere with the mapping or analysis.
Based on all the QC metrics reviewed, no major issues were detected in this dataset using Multiqc. The data were determined to be appropriate for continuing with downstream steps in the RNA-seq analysis pipeline.

RNA-Seq Analysis

First, the required libraries were loaded. Sample names and treatment conditions were assigned based on GEO metadata from GSE288148. All six samples are paired-end RNA-seq from murine CD8+ T cells, comparing Biotin+ and Biotin− conditions (corresponding to tumour-neighbouring and tumour-distal CD8+ T cells).

Low expressed genes were removed, and to account for normalisation in sequencing depth for each of the samples, size factors were estimated to normalise the raw count data.

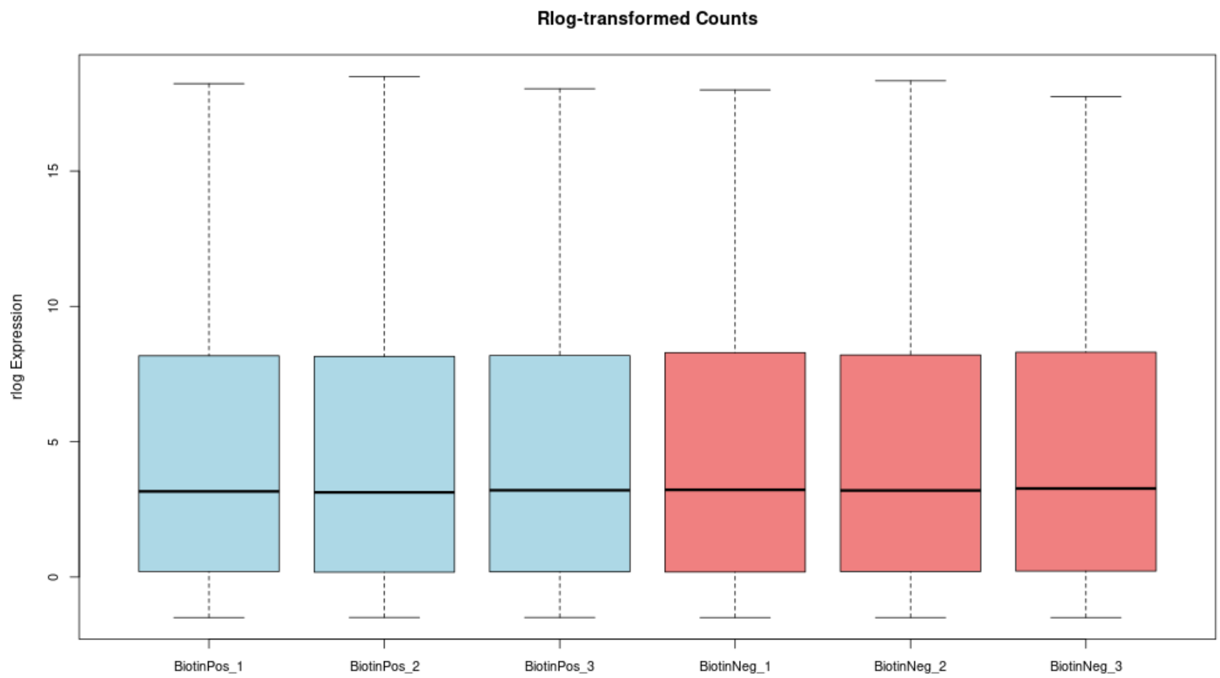

Figure 4: Boxplot of rlog-transformed counts. The transformation stabilises variance across the six samples. Each treatment group (Biotin+/BiotinPos and Biotin−/BiotinNeg) shows consistent expression distributions, with no major outliers observed.

Other normalisation methods (log2, FPM, VST) were explored and showed similar trends. These plots are available in the code for the RNA seq analysis, which is the R markdown (Analysis.Rmd). Since this study involves only six samples, rlog was chosen for visualisations as it performs better with smaller sample sizes (Chipster DESeq2 docs: https://chipster.csc.fi/manual/deseq2-transform.html).

Exploratory Data Analysis

To explore sample relationships and variation in gene expression, Principal Component Analysis (PCA) was performed using rlog-transformed counts. The first principal component (PC1) captured 79% of the variance and clearly separated the two treatment groups: BiotinPos (tumour-neighbouring) and BiotinNeg (distal) (Figure 5). The clustering of each group together suggests consistent expression profiles and minimal variation within groups.

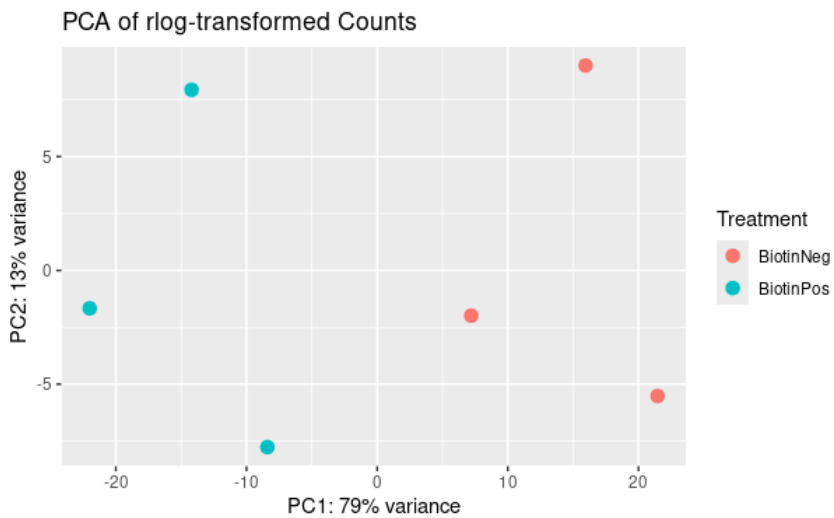

Figure 5: PCA of rlog-transformed expression data. PC1 separates samples based on treatment condition, explaining 79% of the variance. BiotinPos and BiotinNeg samples cluster together within their respective groups.

A heatmap of sample-to-sample distances based on the rlog data was also generated (Figure 6). The clustering pattern in the heatmap is consistent with the PCA results, supporting the interpretation that the main source of variation is the treatment condition.

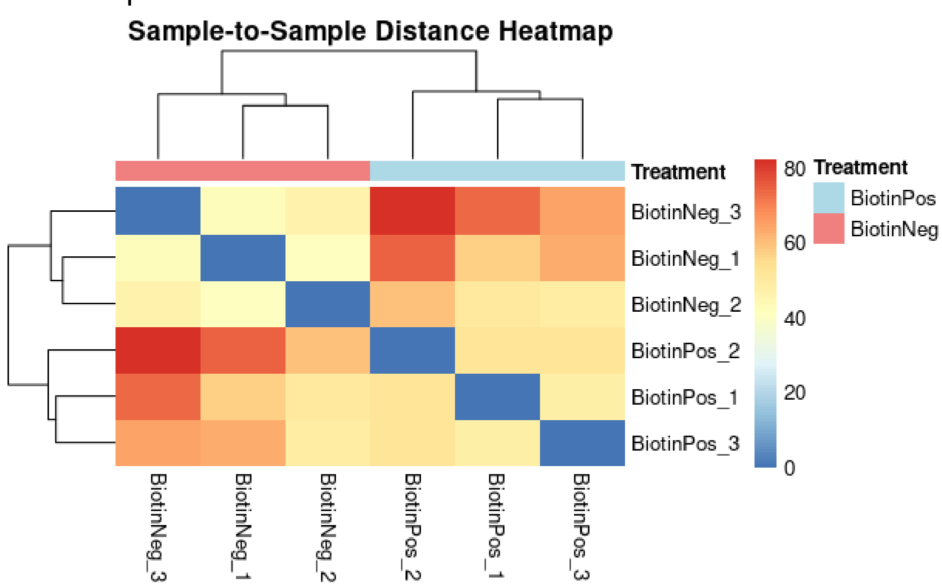

Figure 6: Sample-to-sample distance heatmap using rlog-transformed data. Hierarchical clustering confirms the PCA result, with BiotinPos and BiotinNeg samples forming distinguishable clusters.

Differential Gene Expression Analysis

Differential expression analysis was performed using DESeq2, comparing the BiotinPos (tumour-proximal) group to the BiotinNeg (tumour-distal) group. The BiotinNeg group was set as the reference level, so all log2 fold change values reflect expression in BiotinPos relative to BiotinNeg.

Significant genes were identified using an adjusted p-value threshold of 0.05, which accounts for multiple testing using the Benjamini-Hochberg method. Genes were then ranked by the absolute value of the test statistic, which explains both the size and reliability of the fold change. This was chosen to prioritise genes that showed the most consistent and strong differences in expression between groups.

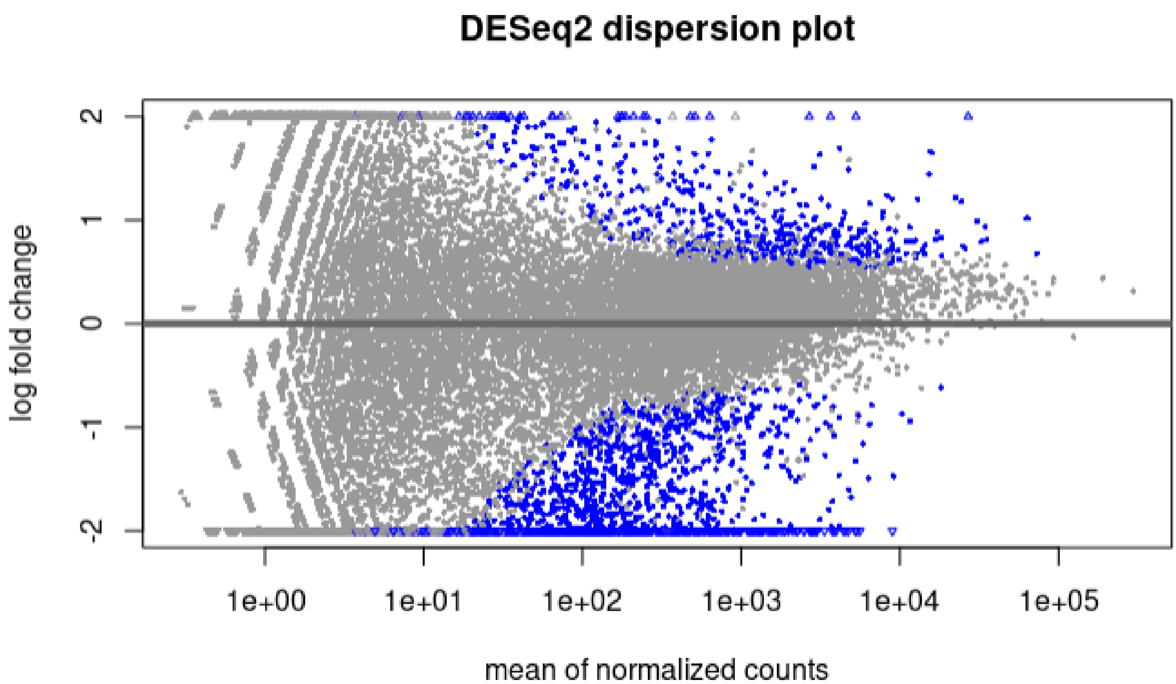

Figure 7: MA plot showing log2 fold changes (y-axis) across the range of mean expression values (x-axis). Blue points represent genes with adjusted p-value < 0.05.

Top 10 Differentially Expressed Genes

The table below lists the top 10 DEGs, ranked by pvalue. Each gene has a strong statistical signal, with adjusted p-values below 0.05, and significant log2 fold changes.

Table 1: Top 10 DEGs ranked by their p-values.

Ensembl_ID	Gene	log2FoldChange	pvalue	padj
ENSMUSG00000019929	Dcn	-2.53377	5.02E-26	9.23E-22
ENSMUSG00000001506	Col1a1	-2.93116	3.21E-20	2.04E-16
ENSMUSG00000015441	Gzmf	3.070367	4.19E-20	2.04E-16
ENSMUSG00000016494	Cd34	-2.55154	4.44E-20	2.04E-16
ENSMUSG00000029781	Fkbp9	-2.65981	6.08E-19	2.24E-15
ENSMUSG00000027962	Vcam1	-2.20845	1.17E-18	3.59E-15
ENSMUSG00000001435	Col18a1	-2.40781	4.73E-18	1.24E-14
ENSMUSG00000031375	Bgn	-2.31576	8.69E-17	2.00E-13
ENSMUSG00000000782	Tcf7	-1.79433	1.16E-16	2.37E-13
ENSMUSG00000001036	Epn2	-2.48924	3.84E-16	7.06E-13

The log2 fold change shows the direction and size of differential expression. For example, a value of -2.5 means the gene is around 5.6x less expressed in BiotinPos samples than in BiotinNeg. The adjusted p-value indicates statistical significance after correcting for multiple comparisons. All genes listed have strong statistical support, and most are biologically relevant based on descriptions in the annotation table.

A volcano plot was also created to highlight functionally important genes reported in the source study (Lou et al., 2025). All candidate genes mentioned in the paper were present in our differential expression results, supporting the validity of the analysis

Functional Enrichment Analysis

The Wald test statistic from DESeq2 was used to rank genes for GSEA, as it reflects both the direction and the reliability of expression changes. This was preferred over log2 fold change alone, since large fold changes with high variability might rank too highly if not adjusted.

The fgsea algorithm was then used to test whether members of predefined Hallmark gene sets tend to cluster at the top (upregulated in Biotin+) or bottom (downregulated) of the ranked gene list.

Significantly upregulated pathways in the Biotin+ group included Oxidative Phosphorylation, MYC Targets V1, and E2F Targets, suggesting increased metabolic and proliferative activity. The most downregulated pathways included Epithelial–Mesenchymal Transition, Angiogenesis, and Hypoxia.

These results support the biological differences described in the source publication and demonstrate the utility of GSEA on a Wald-statistic ranked list. Enrichment plots were created using plotEnrichment() from the fgsea package, with ggplot2 used for figure formatting.

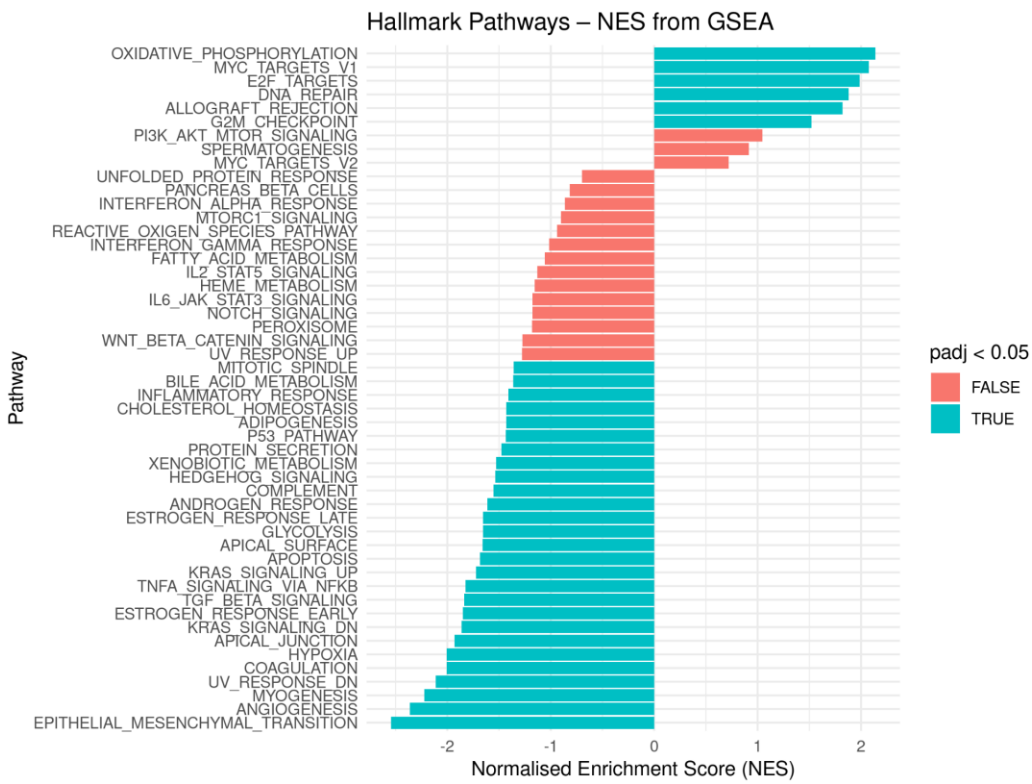

Figure 9: Functional enrichment analysis using Hallmark gene sets. Normalised Enrichment Scores (NES) from fgsea are shown for all tested pathways. Blue bars indicate significantly enriched pathways (adjusted p-value < 0.05).